

# Course Parts List (Main Components)

This list summarizes the main hardware used across the Zephyr ESP32-S3 course.

## Core Required Parts

| Part                      | Purpose in Course                              | Interface  | Specs Found in Repo                                                                                                 | Typical Wiring in Course                                                                              |
|---------------------------|------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| ESP32-S3 DevKitC          | Main target board for all apps                 | MCU board  | Built-in Wi-Fi + BLE support in Zephyr; integrated USB JTAG for debug over single USB cable                         | Used as host for all peripherals                                                                      |
| SSD1306 OLED module       | Display output in diagnostics and sensor demos | I2C        | 0.96 inch, 128x64, default address <code>0x3C</code> (some panels <code>0x3D</code> )                               | <code>SDA -&gt; GPIO1, SCL -&gt; GPIO2, VCC -&gt; 3V3, GND -&gt; GND</code>                           |
| SHT40 (SHT4x) sensor      | Temperature + humidity readings                | I2C        | Default address <code>0x44</code> , alternate <code>0x45</code>                                                     | <code>SDA -&gt; GPIO1, SCL -&gt; GPIO2, VCC -&gt; 3V3, GND -&gt; GND</code>                           |
| MCP3008 ADC               | Analog input acquisition and SPI demo          | SPI        | 8-channel, 10-bit ADC; demo uses SPI mode 0 at 1 MHz                                                                | <code>SCLK -&gt; GPIO12, MOSI -&gt; GPIO11, MISO -&gt; GPIO13, CS -&gt; GPIO17, shared 3V3/GND</code> |
| SG90 hobby servo          | PWM motion control demo                        | PWM (LEDC) | 3-wire servo; slide notes list ~9 g and ~1.5 kg.cm torque; demo timing uses 20 ms period (50 Hz), ~1.0-2.0 ms pulse | <code>Signal -&gt; GPIO15, VCC -&gt; 5V, GND -&gt; GND</code> (shared ground with board)              |
| User button               | Input/interrupt/debounce demos                 | GPIO       | Active-low with pull-up in course overlays                                                                          | <code>GPIO10</code>                                                                                   |
| 2x LEDs                   | Blinky/threading/queue demos and status output | GPIO       | Active-high in course apps; use a series current-limiting resistor per LED                                          | <code>LED0 -&gt; GPIO8, LED1 -&gt; GPIO9</code>                                                       |
| Breadboard + jumper wires | Prototyping interconnect                       | N/A        | Required for external sensors/actuators                                                                             | Used for all external modules                                                                         |
| USB data cable            | Flashing, monitor, debug                       | USB        | Single cable used for flash/serial/JTAG workflow                                                                    | USB connection to ESP32-S3 board                                                                      |

# Common Supporting Parts

| Part                              | Purpose                                        | Specs Found in Repo                                                                      |
|-----------------------------------|------------------------------------------------|------------------------------------------------------------------------------------------|
| Potentiometer (for MCP3008 demos) | Provides variable analog input to ADC channels | Used as analog source in SPI/ADC lecture narrative                                       |
| External 5V supply (servo)        | Powers servo safely under load                 | Course notes recommend external servo supply; share grounds with ESP32-S3                |
| 2x LED current-limiting resistors | Protect GPIO/LED when using external LEDs      | One resistor in series with each LED; typical hobby values are 220 ohm to 1 kohm at 3.3V |